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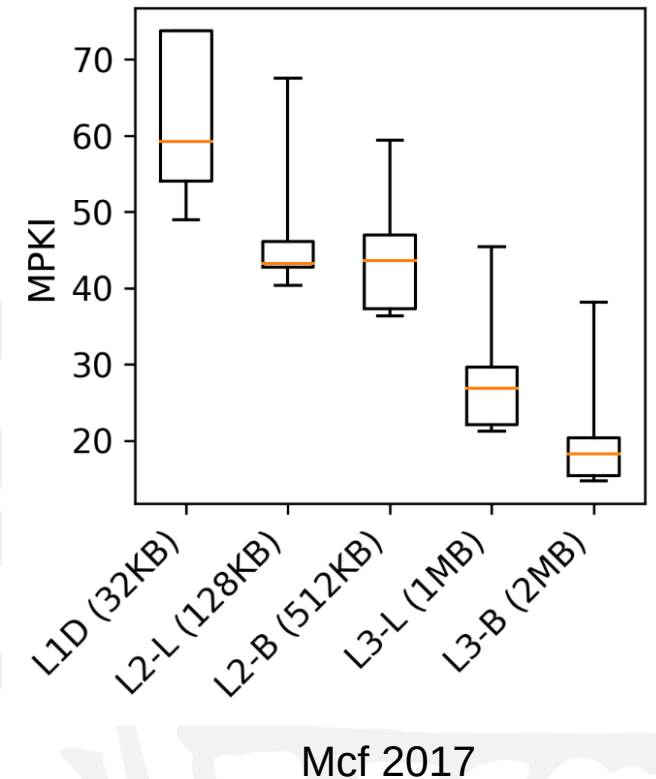
Architecturally-independent and time-based characterization of SPEC CPU 2017

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Problem with μ -architecture dependent characterization

- System specific characterization (using HW perf counters):
 - ✓ simple and fast
 - ✗ Tie the results to a fixed micro-architecture configuration
 - ✗ Results are not applicable to other systems with different configurations
 - ✗ Complex micro-architectural interactions hindering insight
- The figure on right shows MPKI for mcf 2017 from a cycle accurate simulation.
 - Different micro-architectural components (different prefetchers, big (B) and little (L) cores) cause large variation in cache behaviour

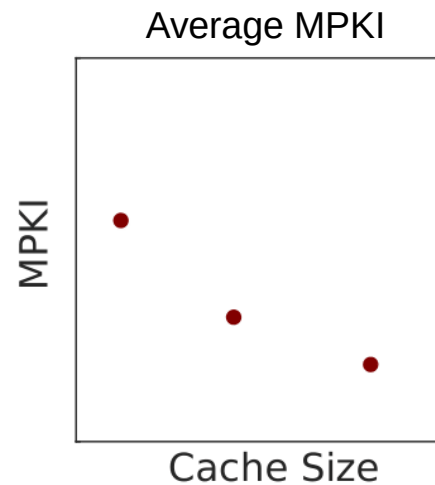


μ -architecture dependent characterization is tied to fix configurations with little flexibility and insight



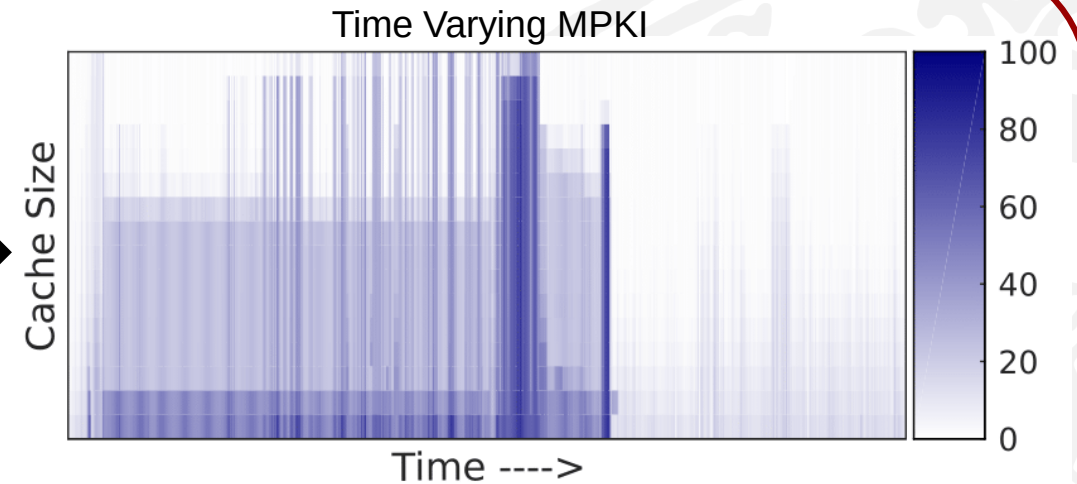
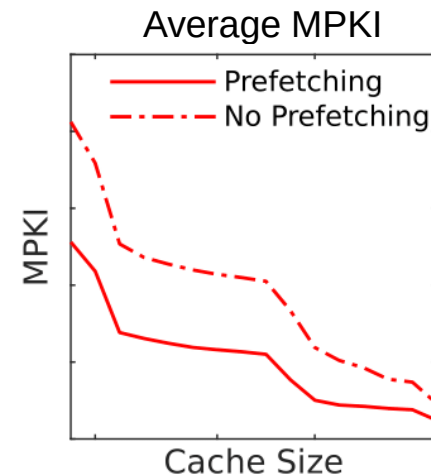
Difference with Previous Works

Previous work



Analyzed aggregate statistics from HW performance counters (fixed cache size and hierarchy)

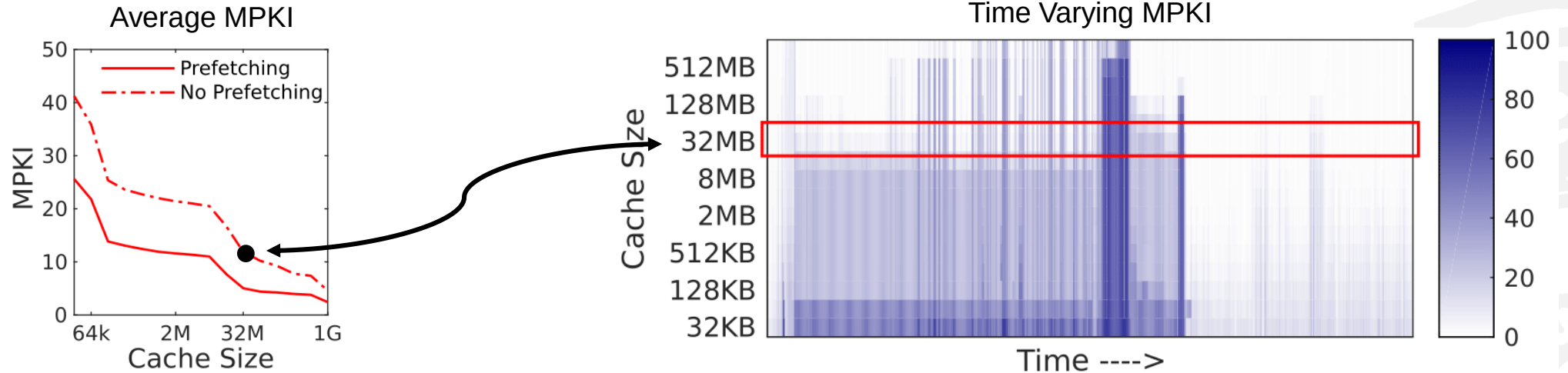
Our Contribution



1. Thorough study of MPKI over varying cache sizes, with and without prefetching
2. Study of application behaviour over time, resulting in new perspective and quantification of memory phases



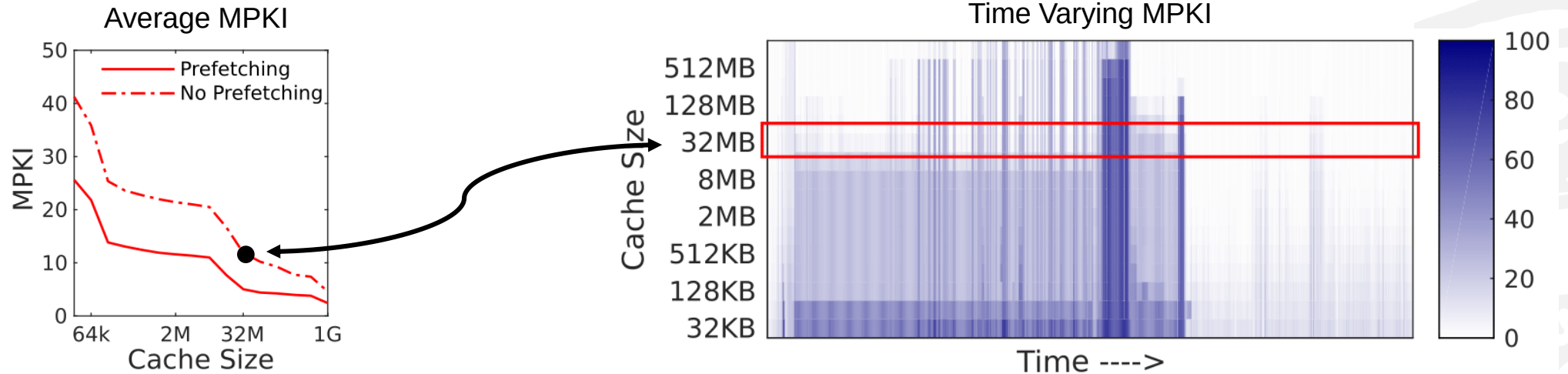
Average & Time Varying MPKI



- Average MPKI of full program execution
- Shows variation of average behaviour with different cache sizes and effect of prefetching
- Shows variation of MPKI during the program execution as a function of both time and cache sizes
- Despite average MPKI of 12 at 32MB cache, certain phases of execution have $MPKI > 50$



Average & Time Varying MPKI



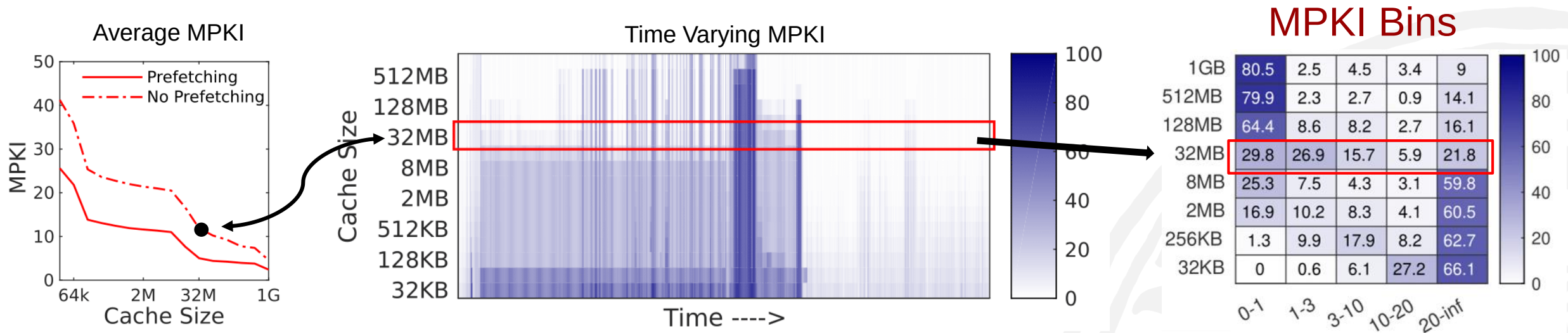
How to derive more Insights from time based MPKI

- What is the frequency of occurrence of phases with MPKI > 10 with 32MB cache?
- How to quantify what percentage of execution has MPKI > 20?
- Is there a way to classify the execution into memory phases based on the MPKI each phase experience?
- e.g. X% of execution in MPKI < 3 range etc. ?





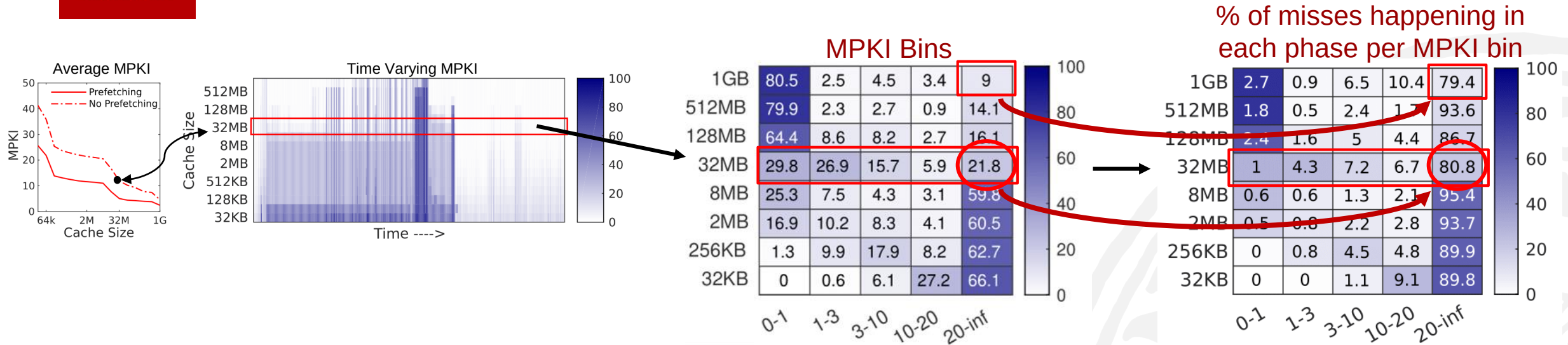
Quantifying and Classifying Memory Phases



- Time based plot shows that certain phases experience >20 MPKI, MPKI bins helps quantify that almost 22% of execution experiences >20 MPKI
- Despite the average MPKI of 12, 55% of execution has < 3 MPKI at 32MB cache
 - **Average MPKI tells very little of what is actually happening!**
- MPKI bins classify phases of execution based on the MPKI of each region (x-axis shows MPKI range)
- MPKI bins show the percent of program execution experiencing a certain MPKI
- **With 32MB cache, almost 22% of execution has MPKI >20, but how bad is this phase?**
- **What is the portion of misses that happen when execution is in this memory intensive phase?**



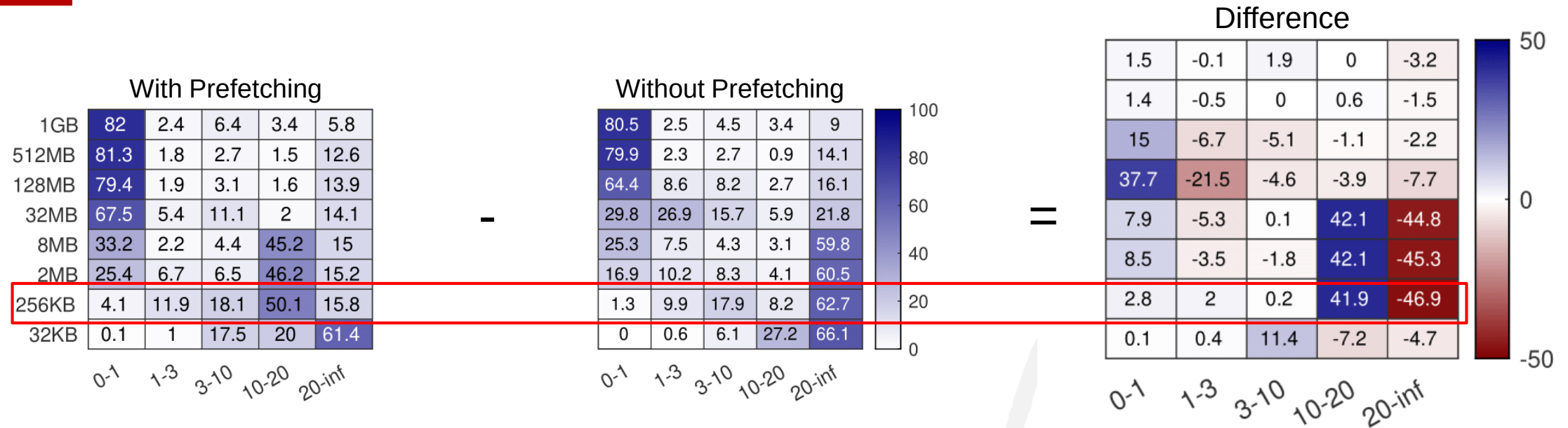
Characterizing Misses Across Different MPKI Bins



- With 32MB cache 22% of the execution is experiencing 81% of the total misses!
- Almost 9% of execution with 1GB cache has MPKI > 20, but accounts for almost 80% of the total cache misses
- Phases with high MPKI account for disproportionately large number of misses
- For overall program optimization, memory bound phases must be optimized



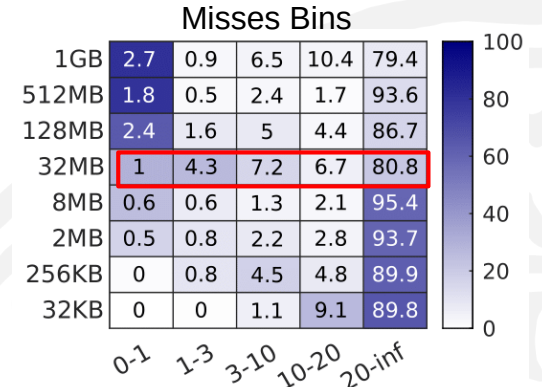
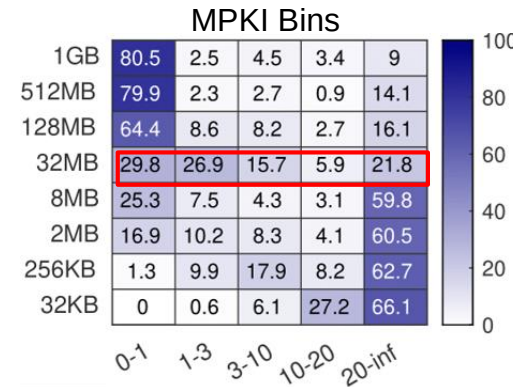
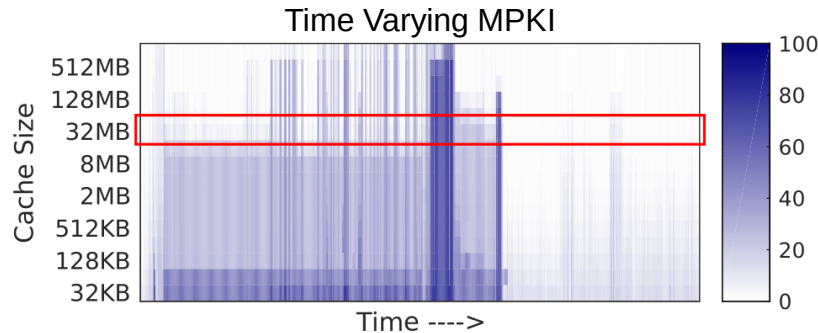
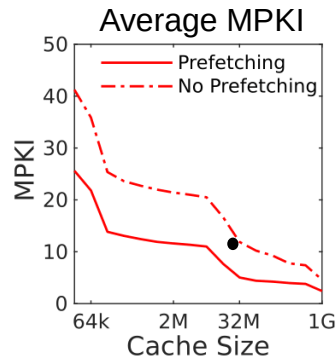
Visualizing the Effect of Prefetching



- MPKI bins help us understand the effect of prefetching
- The difference figure shows MPKI bins with difference in MPKI between with and without prefetching
- With 256 KB cache, prefetching moved almost 47% of execution from >20 MPKI bin to <20 MPKI bin
- MPKI Bins enable compact way of understanding and quantifying effect of prefetching



Complete Characterization



Our characterization provides:

- Average MPKI of the program
- MPKI varying over time and cache size
- Histogram of program execution for different bins of MPKI
- Portion of cache misses that occur at each MPKI bin
- Visualization of the benefits of prefetching

Full data available at: <http://urn.kb.se/resolve?urn=urn:nbn:se:uu:diva-417174>